The Web of Bonds: How the DICTONIC Bonds Hand Over



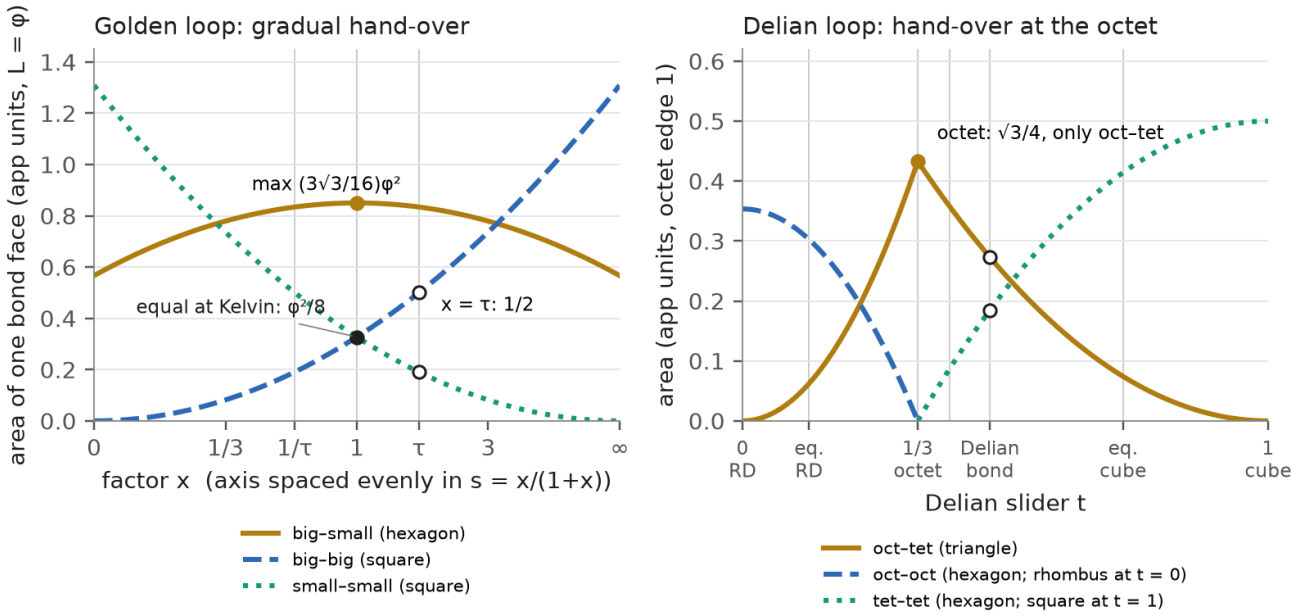
Theorem: both DICTONIC loops are weighted Voronoi (power) diagrams of one fixed crowd of points of two kinds, with one weight slid between the kinds; the bonds are the edges of the dual weighted-Delaunay web; the bond between unlike pieces never breaks inside a loop, while the two like bonds hand over, gradually in the golden loop and all at once, at the octet, in the Delian loop

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Paper #52 · Version 2 · 11 November 2026 (Version 1: 25 October 2026) · DICTO DOI: 10.5281/zenodo.23629252 · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/28cfdde/studies/2026-10-25-web-of-bonds, …/studies/2026-10-25-physical-anchors and …/studies/2026-10-25-scale-anomalies (research branch, commit 28cfdde; also attached to this record)
The ideas are DICTO's: the DICTONIC SOLIDS and the DICTONIC BONDS and their names, and the web of bonds, a network of bonds that "should create a web of lattices that change with the rhythm of the loop" (DICTO, 25 October 2026); computation, exact proofs and drafting with Claude (Anthropic).

Version 2 adds the permutohedron reading (no result changes): in the golden network the weight difference of the Main theorem, $(\varphi^2/4)(x-1)/(x+1)$, is $(L/2)^2$ times $(g-m)/(g+m)$, where g : m : g are the gaps between the four values whose permutohedron is the cell at the lattice points ($x = g/m$). The walk slides the spacing of the values; the cells are those of #90, Proposition 6 (to appear in DICTO, The Permutohedron Paradigm, Chapter 3).

Abstract. DICTO's two loops of space-filling pairs, the golden loop of paper #48 and the Delian loop of paper #51, each slide one pair of solids through a whole family of shapes. Draw one line, a bond, between every two pieces that share a face, and the lines make a web. We show that each loop is really one fixed crowd of points of two kinds, the piece centres, and that the only thing that moves is one number, a weight given to one kind against the other. Every piece is the region nearer (in this weighted sense) to its own point than to any other. Then the web is exactly the web of the classical weighted Delaunay construction. As the weight slides, the bond between unlike pieces never breaks. The two like bonds hand over: in the golden loop gradually, both alive all the way and equal at Kelvin's cell; in the Delian loop all at once, at the octet, where every rhombus of a hidden rhombic dodecahedron trades its long diagonal for its short one. Read physically, the weight is a ball radius squared: the golden loop is the caesium chloride structure, the Delian loop the fluorite structure, and the range where the balls can touch is bounded exactly by Pauling's classical radius-ratio limits (the same contact condition as Pauling's).
Main theorem. For every x in $[0, \infty]$ the golden network is the power diagram of the body-centred cubic sites with weight difference $(\varphi^2/4)(x-1)/(x+1)$; for every t in $[0, 1]$ the Delian network is the power diagram of the fluorite sites with weight difference $(3/8)(2t-1)$. The web is the 1-skeleton of the dual regular subdivision, with exact bond areas. At the ends, at Kelvin's cell and at the octet these are classical Voronoi or Delaunay tessellations; the theorem gives the whole family between them. In the golden loop all three bond types live on the whole open loop, with constant face lattice; the two like bonds have area ratio x^2 and are equal only at $x = 1$. In the Delian loop the unlike bond

lives exactly on (0, 1), oct–oct exactly on [0, 1/3), tet–tet exactly on (1/3, 1]; t = 1/3, the octet, is the only parameter at which both pieces exist and only one bond type is present. **Theorem 2 (the balls):** the weight is a squared radius; with unlike neighbours touching, ρ = (x + 2)/(2x + 1) and ρ = t/(1 − t), and the touching windows end exactly at Pauling's limits √3 − 1 and √(3/2) − 1 (Pauling's own contact condition, rederived). All geometric results are exact (371 + 61 checks, none failed; the 24 physical checks of [S3] are estimates, Appendix B).



DICTO

Figure 1. The exact area of one bond face along each loop (Main theorem; Table 1). Left, golden loop (axis evenly spaced in s = x/(1 + x), so the whole loop 0 ≤ x ≤ ∞ fits): the unlike bond (gold) never breaks and peaks at Kelvin; the two like bonds trade size, equal at Kelvin (φ²/8), and big–big 1/2 against small–small 1/(2φ²) at x = τ. Right, Delian loop: oct–oct (blue) dies exactly where tet–tet (green) is born, at the octet t = 1/3, where the unlike bond peaks, with a corner, at the octet triangle √3/4. Open circles: x = τ and the Delian bond.

1. Where it started

> *DICTO's DICTONIC world in Kaleidohedra shows a web: one line for every face two pieces share. DICTO's idea was that this web should itself be a lattice that changes with the rhythm of the loop. We asked what that web is, exactly.*

Paper #48 [1] cut the gaps between icosahedra at the golden point and placed the resulting pair, the Icosaxes and the Unborn Axes, in a one-parameter family T(x), T(1/x) of truncated octahedra (Koca's orbit [10]). Paper #51 [2] cut the octet by one sliding plane per tetrahedron–octahedron pair and found the Delian bond, Samson and Delilah. DICTO called the solids the DICTONIC SOLIDS and the pairs the DICTONIC BONDS, and in the app each family runs as a loop. On 25 October 2026 DICTO's idea was that the network of bonds should create a web of lattices that change with the rhythm of the loop. The app draws that web (Figure 5): one line per shared face, thicker for a larger face. Float runs on a small network suggested that bonds trade places along each loop. Two exact studies [S1, S2] answer what the web is, and a third [S3] tests the physical reading (§6); this paper reports them.

2. The two loops

> *Both loops are pairs of pieces that fill space. Each piece sits on a point; the points never move. Only the shapes change.*

Golden loop. T(x) is the convex hull of all permutations of (±(1 + x), ±x, 0), scaled so that its squares lie at distance L/2 from its centre, with L = φ (paper #48, Theorem 4). T(x) sits on the big sites B = φℤ³, T(1/x) on the small sites S = φℤ³ + (φ/2)(1, 1, 1), for x from 0 to ∞. We write s = x/(1 + x) in [0, 1]: then T(x) is the set of signed permutations of (1, s, 0)·L/2, and T(1/x) the same with 1 − s in place of s. At x = 1 both are Kelvin's truncated octahedron [20]; at x = τ (= φ, the name of the stop in paper #48 and the app) they are the Icosaxes and the Unborn Axes. "Big" and "small" name the two kinds of site (the kinds of #48, the Icosaxes big); for x < 1 the big piece is the smaller of the two, and the swap x ↔ 1/x exchanges them.

Delian loop. The pieces of paper #51 at slider t in [0, 1], with a = 3(1 − t)/2: the tetrahedron pieces sit on Q = (ℤ + ½)³/√2, the octahedron pieces on O = {n ∈ ℤ³ : $n_1 + n_2 + n_3$ odd}/√2, octet edge 1. At t = 0 the octahedron pieces are Kepler's rhombic dodecahedra, at t = 1/3 the octet, at t = 1 the tetrahedron pieces are cubes; in between lie the two equal-edged members, the half-and-half member and the Delian bond (Samson and Delilah).

Power diagram. Given sites c_i with weights w_i, the power cell of c_i is the set of points p with $|p - c_i|^2 - w_i \le |p - c_j|^2 - w_j$ for all j: a Voronoi cell in which each site is given a head start [3, 4]. Two cells meet in a plane perpendicular to the line between their sites (the power bisector). The cells are convex and fill space. Their dual, the weighted (regular) Delaunay subdivision, has an edge between two sites exactly when their cells share a face [3].

3. Main theorem

> *Each loop is one fixed crowd of points with one weight slid between the two kinds. The web of bonds is the classical dual web. The unlike bond never breaks; the like bonds hand over.*

The classical members. Some members of both loops are classical. Every Delaunay tessellation is the power diagram of its own circumspheres [3, 23]. The octet (t = 1/3) is the Delaunay tessellation of the face-centred cubic lattice; its circumspheres sit on the octahedron centres ($r^2 = 1/2$) and the tetrahedron centres ($r^2 = 3/8$), a weight difference $w_Q - w_O = -1/8$. The octahedron–cuboctahedron honeycomb at x = 0 and x = ∞ is likewise a power diagram of its circumspheres ($r^2 = φ^2/4$ and $φ^2/2$), a weight difference $-φ^2/4$ between octahedron and cuboctahedron. Kelvin's cells (x = 1), the rhombic dodecahedra (t = 0) and the cubes (t = 1) are ordinary Voronoi diagrams (of the surviving kind of site alone at t = 0 and t = 1). These are exactly the weights of the theorem below at those points. What the theorem adds is the whole family between these classical members.

> **Main theorem.** (i) For every x ∈ [0, ∞] the golden network is the power diagram of the body-centred cubic sites B ∪ S with $w_B - w_S = (φ^2/4)(x - 1)/(x + 1) = (L/2)^2(2s - 1)$. For every t ∈ [0, 1] the Delian network is the power diagram of Q ∪ O with $w_Q - w_O = (3/8)(2t - 1)$. At t = 0 the Q cells are single points, at t = 1 the O cells are single points.
> (ii) The web of bonds is the 1-skeleton of the dual regular subdivision, and the bond faces and their areas are those of Table 1.
> (iii) *Golden: gradual hand-over.* For 0 < x < ∞ all three bond types are present and the face lattice is constant. Big–big rises strictly from 0 and small–small falls strictly to 0; their ratio

is x^2, the diagonals of the two square faces (sL and (1 − s)L) add up to the lattice spacing φ, and the two areas are equal only at x = 1. The whole picture is symmetric under x ↔ 1/x. *Delian: hand-over at the octet.* Oct–tet is present exactly on (0, 1), oct–oct exactly on [0, 1/3), tet–tet exactly on (1/3, 1]. t = 1/3 is the only parameter at which both pieces exist and every face is of one bond type (there the web is bipartite). The ends t = 0 and t = 1 carry one bond type only because one piece has shrunk to a point. The golden loop never has a single bond type: at x = 0 there are two (big–small and small–small), at x = ∞ two (big–small and big–big).

Loop	Bond	Face	Area (app units)	Alive	Shape of the curve
golden (L = φ)	big–small	hexagon (triangle at x = 0, ∞)	$(\sqrt{3}/8)\varphi^2\cdot(x^2 + 4x + 1)/(1 + x)^2$	whole loop	rises on (0, 1), falls on (1, ∞); max $(3\sqrt{3}/16)\varphi^2 \approx 0.8502$ at Kelvin; $(\sqrt{3}/8)\varphi^2 \approx 0.5668$ at both ends
	big–big	square, diagonal sL	$\varphi^2x^2/(2(1 + x)^2)$	born at x = 0^+	strictly increasing, → $\varphi^2/2$ as x → ∞
	small–small	square, diagonal (1 − s)L	$\varphi^2/(2(1 + x)^2)$	dies as x → ∞	strictly decreasing from $\varphi^2/2$ at x = 0
Delian (octet edge 1)	oct–tet	triangle	$(9\sqrt{3}/4)t^2$ for t ≤ 1/3; $(9\sqrt{3}/16)(1 - t)^2$ for t ≥ 1/3	exactly (0, 1)	rises to √3/4 (the octet triangle) at 1/3, then falls; a corner at 1/3, slopes $3\sqrt{3}/2$ and $-3\sqrt{3}/4$
	oct–oct	hexagon (rhombus at t = 0)	$(\sqrt{2}/4)(1 - 9t^2)$	exactly [0, 1/3)	strictly decreasing from √2/4 (the rhombic dodecahedron's rhombus)
	tet–tet	hexagon (square at t = 1)	$(3t - 1)(5 - 3t)/8 = (1 - a^2)/2$	exactly (1/3, 1]	strictly increasing to 1/2 (the cube's face)

Table 1. The bond faces and their exact areas. Sample values: at Kelvin (x = 1) big–big = small–small = $\varphi^2/8 \approx 0.32725$; at x = τ big–big = 1/2, small–small = $1/(2\varphi^2) \approx 0.19098$, big–small = $(\sqrt{3} + 3\sqrt{15})/16 \approx 0.8344$; at the Delian bond oct–tet = $2^{1/3}\sqrt{3}/8 \approx 0.2728$ and tet–tet = $1/2 - 2^{1/3}/4 \approx 0.1850$.

> *Why: choose the weights so that every flat face of every piece is exactly the plane of equal weighted distance between the two points it separates. Then each piece is its point's weighted cell, and the rest is the classical duality.*

Proof. (i) With the weights in (i), every facet plane of every piece of the app is the power bisector of the two sites that share it, for every s ∈ (0, 1) and every t in each of the regimes (0, 1/3) and (1/3, 1), as an identity in the parameter [S1, A2]. Every corner of the piece satisfies every other power inequality against every site within a fixed distance, with slack ≥ 0 on the whole open regime: the slacks are linear in s or t and have no root inside [S1, A1]. The facet planes' own vertex enumeration gives back exactly the piece's corners, and their normals sum to 0 with rank 3, so they bound a bounded region; together with A1 this is piece = power cell. A uniform distance bound shows that sites farther away cannot cut the cell [S1, A7]: every corner lies within ρ of its site, $\rho^2 \le 2$ in units of $(L/2)^2$ (golden) and $\rho^2 \le 1$ in units where the cube at t = 1 has edge 1 (Delian), and with the site lists out to D = 4 and D = 3 in those units the bound $(D - \rho)^2 \ge \rho^2 + |\Delta|$ holds for every parameter ($(4 - \sqrt{2})^2 \approx 6.69 \ge 2 + 1$; $(3 - 1)^2 = 4 \ge 1 + 3/4$). The regime boundaries and the ends x = 0, ∞, t = 0, 1/3, 1 are checked by building the power cells from scratch among the fixed sites, with no use of the app's pieces, in exact arithmetic, and comparing with the app's pieces; at t = 0 and t = 1 the vanishing kind's cell is a single point [S1, B1]. At t = 1/3 and x = 0, ∞ the identity also follows from the circumsphere description of a Delaunay tessellation (above), an independent check. (ii) Power diagrams are dual to regular subdivisions [3], so two sites are bonded exactly when

they span an edge of a dual cell. The face of each bond is computed in closed form [S1, A3] and checked at every sample, rational and irrational [S1, B2], and against the app's own web to 10^{-9} [S1, C2]. (iii) The monotonicity, birth and death points and the corner at 1/3 follow from the closed forms [S1, A6]; the ratio of the like bonds is $s^2/(1-s)^2 = x^2$, so they are equal only at x = 1; the square diagonals are sL and (1 − s)L, which add to L = φ. The swap: $s \leftrightarrow 1 - s$ flips the sign of the weight, and the translation by (φ/2)(1, 1, 1) exchanges B and S, so the diagram at 1/x is the diagram at x translated with big and small exchanged; hence big–big(1/x) = small–small(x) and big–small is unchanged. The bond types present in each regime are read off the incidences of the dual cells (Corollary 3) [S1, B6]. ∎

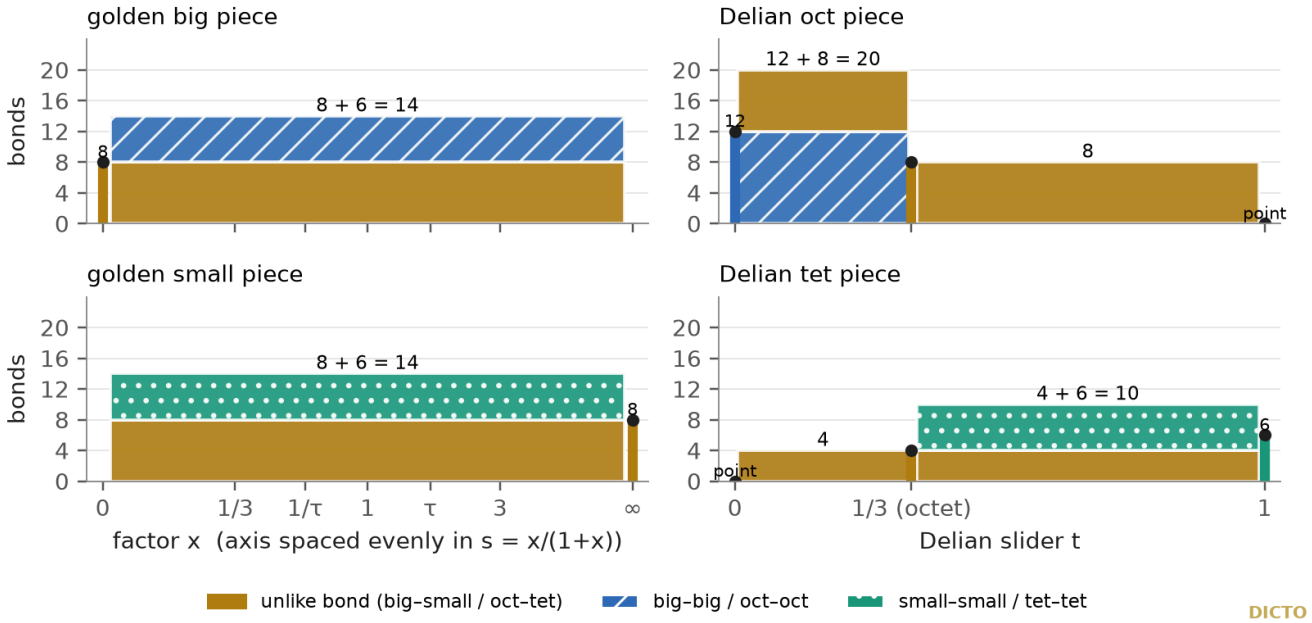


Figure 2. Bonds per piece along each loop (Corollary 1). Bars: open intervals; dots: single parameter values. The golden pieces keep 8 unlike bonds throughout and 6 like ones except at one end. The Delian octahedron piece drops from 20 bonds to 8 at the octet, and the tetrahedron piece rises from 4 to 10; at the octet both pieces have only unlike bonds.

4. Corollaries

> *The theorem tells how many bonds each piece has, where the unlike bond is strongest, and what the hidden dual lattice looks like at each moment of the loop.*

Corollary 1 (bonds per piece). Exact counts [S1, B3, C4]:

	x = 0	0 < x < ∞	x = ∞
golden big	8 big–small (octahedron)	14 = 8 big–small + 6 big–big	14 = 8 + 6 (cuboctahedron)
golden small	14 = 8 + 6 small–small (cuboctahedron)	14 = 8 big–small + 6 small–small	8 (octahedron)

	t = 0	0 < t < 1/3	t = 1/3	1/3 < t < 1	t = 1
Delian oct	12 oct–oct (rhombic dodecahedron)	20 = 12 oct–oct + 8 oct–tet	8 oct–tet	8 oct–tet	none (point)
Delian tet	none (point)	4 oct–tet	4 oct–tet	10 = 4 oct–tet + 6 tet–tet	6 tet–tet (cube)

Each golden piece's 6 squares always bond to its own kind: big–big for a big piece, small–small for a small one. What trades between the like bonds is their size, not the pieces they

join; at an end, one kind's squares shrink to points.

Corollary 2 (the unlike bond's peak). The unlike bond is largest at Kelvin's cell in the golden loop, smoothly, with area $(3\sqrt{3}/16)\varphi^2$, and at the octet in the Delian loop, with a corner, with area $\sqrt{3}/4$.

Corollary 3 (the hidden dual lattice). The dual weighted-Delaunay cells are [S1, B6, B8] (volumes in units of $(L/2)^3$ for the golden loop and, for the Delian loop, of the cube that the tetrahedron piece becomes at t = 1 (edge $1/\sqrt{2}$ at octet edge 1); exact):

Loop, range	Dual cells	Volumes
golden, 0 < x < ∞	disphenoid tetrahedra, 2 big + 2 small sites: one subdivision throughout, **no flip**	2/3
golden, x = 0 and x = ∞	square bipyramids (2 + 4 sites)	8/3
Delian, t < 1/3	octahedra (6 O) and tetrahedra (3 O + 1 Q)	4/3 and 1/12
Delian, t = 1/3	rhombic dodecahedra (6 O + 8 Q)	2
Delian, t > 1/3	cubes (8 Q) and square pyramids (1 O + 4 Q)	1 and 1/6

At t = 1/3 each rhombus of the dual rhombic dodecahedra trades its long diagonal (an oct–oct bond) for its short one (a tet–tet bond): a single flip (Figure 3). At t = 0 the dual is the octet itself; at t = 1 it is cubes.

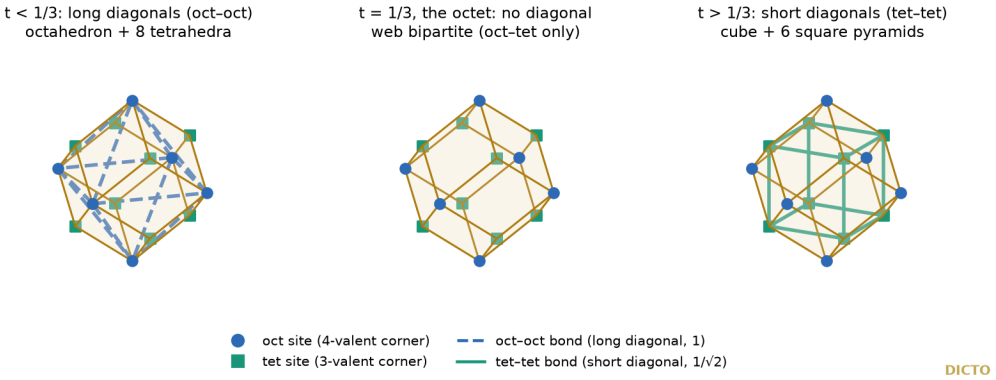


Figure 3. The hand-over at the octet, seen in the dual. One dual cell at t = 1/3 is a rhombic dodecahedron whose 4-valent corners are oct sites (blue) and whose 3-valent corners are tet sites (green). Before the octet (left) every rhombus is split by its long diagonal, an oct–oct bond of length 1, and the cell falls into an octahedron and 8 tetrahedra. At the octet (middle) no diagonal is drawn and every bond joins unlike sites. After it (right) every rhombus is split by its short diagonal, a tet–tet bond of length $1/\sqrt{2}$, and the cell falls into a cube and 6 square pyramids. Octet edge 1.

Corollary 4 (the weight line). For fixed sites, the weights that give one and the same dual subdivision form the cones of the secondary fan [5, 6], its chambers being the triangulations; for periodic point sets see Alexeev [24, §5.12]. Each loop's weight runs along a segment. The golden segment stays inside one chamber except at its ends; the Delian segment lies in two non-maximal cones (its dual cells are not simplices, because the tied weights keep six oct sites co-spherical) that meet exactly once, at t = 1/3, in the cone of the rhombic dodecahedra.

Corollary 5 (the Delian bond). At the Delian bond, $w_Q - w_O = 3/8 - 2^{2/3}/4 \approx -0.02185$: close to, but not at, the unweighted Voronoi point t = 1/2. The ratio of its two bond areas is oct–tet / tet–tet $= 2^{1/3}\sqrt{3} / (4 - 2\cdot 2^{1/3})$.

Remark (the weights in earlier work). These weights agree with paper #51's cube Remark, $w = (x - 1)/(4(x + 1))$ at cube edge 1, and with the octet check of #51's study; the theorem proves them for every parameter, not only at samples. The golden volumes are $(4/3)(1 + u)^3 - 4u^3$ in units of $(L/2)^3$, with $u = s$ for big and $u = 1 - s$ for small, always adding to L^3; the Delian volumes are those of paper #51.

5. The physical reading: balls of two sizes

> *A weight in a power diagram can be read as the square of a ball's radius: the face between two touching balls passes through their point of contact. Read this way, the two loops are two classical crystal structures, and the range where the balls can touch ends exactly at Pauling's radius-ratio limits.*

> **Theorem 2 (the balls).** (a) The radical tessellation of a packing of balls (c_i, r_i) is the power diagram with $w_i = r_i^2 + C$, for any common constant C; a power diagram therefore fixes the radii only up to $r_i^2 \to r_i^2 + C$. (b) The golden sites are the caesium chloride structure (B2, Wyckoff 1a and 1b, lattice constant $a = \varphi$ in app units); the Delian sites, shifted by $(1, 0, 0)/\sqrt{2}$, are the fluorite structure (C1: the oct sites on the face-centred cubic 4a, the tet sites on 8c, all the tetrahedral holes; $a = \sqrt{2}$). (c) If unlike neighbours touch, the radius ratio is $\rho = r_{small}/r_{big} = (x + 2)/(2x + 1) = (2 - s)/(1 + s)$ on the golden loop and $\rho = r_{tet\text{-}site}/r_{oct\text{-}site} = t/(1 - t)$ on the Delian loop; so t is exactly the share of the unlike bond length taken by the tet-site ball. (d) The touching window (unlike balls touching, like balls not overlapping) is $x \in [(\sqrt{3} - 1)/2, \sqrt{3} + 1]$, $\rho \in [\sqrt{3} - 1, (\sqrt{3} + 1)/2]$ on the golden loop, and $t \in [1 - \sqrt{(2/3)}, 1/\sqrt{3}]$, $\rho \in [\sqrt{(3/2)} - 1, (\sqrt{3} + 1)/2]$ on the Delian loop. Its ends are Pauling's radius-ratio limits [9] and their reciprocals: $\sqrt{3} - 1 \approx 0.732$ for 8-coordination (golden lower end; Delian upper end as $1/(\sqrt{3} - 1) = (\sqrt{3} + 1)/2$), and $\sqrt{(3/2)} - 1 \approx 0.225$ for 4-coordination (Delian lower end). (e) Without the touching condition, some non-overlapping ball packing realises every $x \in [0, \infty]$, but only the $t \in [1/6, 2/3]$.

> *Why: the face between two balls is where their weighted distances agree, and that is the power bisector with weight r^2. Touching fixes the sum of the two radii, the weight fixes the difference of their squares, and the two together fix each radius.*

Proof. (a) The radical plane of two balls is the set where $|p - c_i|^2 - r_i^2 = |p - c_j|^2 - r_j^2$, that is, $2p\cdot(c_j - c_i) = |c_j|^2 - |c_i|^2 + w_i - w_j$, which depends on the weights only through their differences [S2, A1a, A1b]; for touching balls it passes through the contact point [S2, A1c] (classical: [4, 7]). (b) Exact coset comparison with the Wyckoff positions [S2, A6]. (c) Let d be the unlike nearest distance and Δ the weight difference of the Main theorem. Then $r_1 + r_2 = d$ and $r_1^2 - r_2^2 = \Delta$ give $r_1 - r_2 = \Delta/d$. Golden: $d = (\sqrt{3}/2)\varphi$ and $\Delta/d^2 = (2s - 1)/3$, so $r_{big}/d = (1 + s)/3$ and $r_{small}/d = (2 - s)/3$. Delian: $d^2 = 3/8$ and $\Delta/d^2 = 2t - 1$, so $r_{tet}/d = t$ [S2, A2, A4]. (d) Golden: the big balls touch ($2r_{big} = a = \varphi$) exactly when $(1 + s)/\sqrt{3} = 1$, at $x = \sqrt{3} + 1$, $\rho = \sqrt{3} - 1$, and the small balls touch at $x = (\sqrt{3} - 1)/2$ by the swap. Delian: the oct balls touch ($2r_O = 1$) at $t = 1 - \sqrt{(2/3)}$, $\rho = \sqrt{(3/2)} - 1$, the tet balls ($2r_Q = 1/\sqrt{2}$) at $t = 1/\sqrt{3}$, $\rho = (\sqrt{3} + 1)/2$ [S2, A2e, A2f, A4c, A4d]. (e) Put $u = r_1 + r_2$; then $r_{1,2} = (u \pm \Delta/u)/2$, both radii grow with u and ρ moves monotonically towards 1, so the feasible u form an interval, nonempty exactly when the larger radius $\sqrt{|\Delta|}$, with the other radius 0, fits. Golden: $|\Delta| \le (a/2)^2 = \varphi^2/4$ always. Delian: $-\Delta \le (1/2)^2$ exactly when $t \ge 1/6$, and $\Delta \le (1/(2\sqrt{2}))^2$ exactly when $t \le 2/3$ [S2, A3, A4f]. And $\sqrt{|$

$\Delta| \le d$ holds for every parameter ($|\Delta| \le (a/2)^2 < d^2$ golden, $|\Delta| \le 3/8 = d^2$ Delian), so unlike balls never overlap at that u. ∎

Corollary 6 (the stops as balls). Under the contact map: golden x = 0 gives ρ = 2 and x = ∞ gives ρ = 1/2, both outside the window, where the only packing has the other kind as points; Kelvin (x = 1) forces ρ = 1, equal balls; x = τ gives ρ = $\sqrt{5}/\varphi^2 = (3\sqrt{5} - 5)/2 \approx 0.8541$, inside. Delian: the octet t = 1/3 gives ρ = 1/2, the Delian bond ρ = $(3/2)\cdot 2^{1/3} - 1 \approx 0.8899$, the unweighted Voronoi point t = 1/2 gives ρ = 1, all inside; t = 0 and t = 1 admit no packing at all; t = 1/6 and t = 2/3 are the end packings (touching face-centred cubic balls with the tet sites as points, and touching simple-cubic balls on the tet sites with the oct sites as points) [S2, A5].

Corollary 7 (an area identity). In the Delian contact model the oct–tet triangle has area $(3\sqrt{3}/2)\cdot r_{oct}^2$ for t ≥ 1/3 and $6\sqrt{3}\cdot r_{tet}^2$ for t ≤ 1/3 [S2, A7]. So compounds with the same oct-site radius have the same oct–tet area: BaF_2, Na_2O and K_2O, all with r_{oct} = 1.42 Å, share 5.239 Å².

Real compounds on the loops

Placing real ionic crystals with Shannon's effective ionic radii [8] (as transcribed by the Imperial College Database of Ionic Radii, which takes all its data from [8]; not checked against the printed page) under the contact model gives Tables 2 and 3 and Figure 4. Every row was checked exactly: the radical cell built with w = r² is the app's piece at that x or t, with the bond counts and areas of Table 1 [S2, B2]. For every caesium chloride type and fluorite compound, "in the window" is the same as $r^+/r^- \ge 0.732$, Pauling's 8-coordination limit; for the antifluorites the same inequality holds with the ion on the 8-coordinated site in the numerator, $r^-/r^+ \ge 0.732$ [S2, B3]. Areas are in app units (golden L = φ; Delian octet edge 1).

Compound	r(cation), CN	r(anion), CN	ρ	x (exact)	big–small	big–big	small–small
CsCl	Cs^+ 1.74, VIII	Cl^- 1.81, VI*	0.9613	188/167 ≈ 1.126	0.8492	0.3671	0.2897
CsBr	Cs^+ 1.74, VIII	Br^- 1.96, VI*	0.8878	109/76 ≈ 1.434	0.8412	0.4544	0.2209
CsI	Cs^+ 1.74, VIII	I^- 2.20, VI*	0.7909	133/64 ≈ 2.078	0.8155	0.5966	0.1382
TlCl	Tl^+ 1.59, VIII	Cl^- 1.81, VI*	0.8785	203/137 ≈ 1.482	0.8396	0.4666	0.2125
TlBr	Tl^+ 1.59, VIII	Br^- 1.96, VI*	0.8112	233/122 ≈ 1.910	0.8225	0.5639	0.1546

Table 2. Caesium chloride type on the golden loop: the larger ion (the anion) on the big site; every row has 8 big–small + 6 big–big bonds per big piece and 8 big–small + 6 small–small per small piece, and lies inside the touching window. Radii in Å. *Shannon gives Cl^-, Br^- and I^- only for CN VI; the CN VI value is used as a proxy for CN VIII.

Compound	oct site (CN 8)	tet site (CN 4)	ρ	t	side of octet	oct–tet	like bond	window
CaF_2	Ca^{2+} 1.12	F^- 1.31	1.1696	0.5391	> 1/3	0.2070	tet–tet 0.2610	in
SrF_2	Sr^{2+} 1.26	F^- 1.31	1.0397	0.5097	> 1/3	0.2342	tet–tet 0.2296	in
BaF_2	Ba^{2+} 1.42	F^- 1.31	0.9225	0.4799	> 1/3	0.2636	tet–tet 0.1956	in
CdF_2	Cd^{2+} 1.10	F^- 1.31	1.1909	0.5436	> 1/3	0.2030	tet–tet 0.2656	in
PbF_2 (β)	Pb^{2+} 1.29	F^- 1.31	1.0155	0.5038	> 1/3	0.2398	tet–tet 0.2231	in
UO_2	U^{4+} 1.00	O^{2-} 1.38	1.3800	0.5798	> 1/3	0.1720	tet–tet 0.3014	**out**
ThO_2	Th^{4+} 1.05	O^{2-} 1.38	1.3143	0.5679	> 1/3	0.1819	tet–tet 0.2900	in
CeO_2	Ce^{4+} 0.97	O^{2-} 1.38	1.4227	0.5872	> 1/3	0.1660	tet–tet 0.3083	**out**
ZrO_2 (cubic)	Zr^{4+} 0.84	O^{2-} 1.38	1.6429	0.6216	> 1/3	0.1395	tet–tet 0.3389	**out**
Li_2O (anti)	O^{2-} 1.42	Li^+ 0.59	0.4155	0.2935	**< 1/3**	0.3358	oct–oct 0.0794	in
Na_2O (anti)	O^{2-} 1.42	Na^+ 0.99	0.6972	0.4108	> 1/3	0.3382	tet–tet 0.1094	in
K_2O (anti)	O^{2-} 1.42	K^+ 1.37	0.9648	0.4910	> 1/3	0.2524	tet–tet 0.2086	in

Table 3. Fluorite and antifluorite on the Delian loop (radii in Å; oct site 8-coordinated, tet site 4-coordinated). In fluorite the cation sits on the oct sites; in antifluorite (Li_2O, Na_2O, K_2O) the anion does. Caution on names: the app's "oct" pieces sit on the 8-coordinated face-centred cubic sites, its "tet" pieces on the 4-coordinated tetrahedral holes. β-PbF_2 is the fluorite form, metastable at ambient (α is orthorhombic); placement is for β.

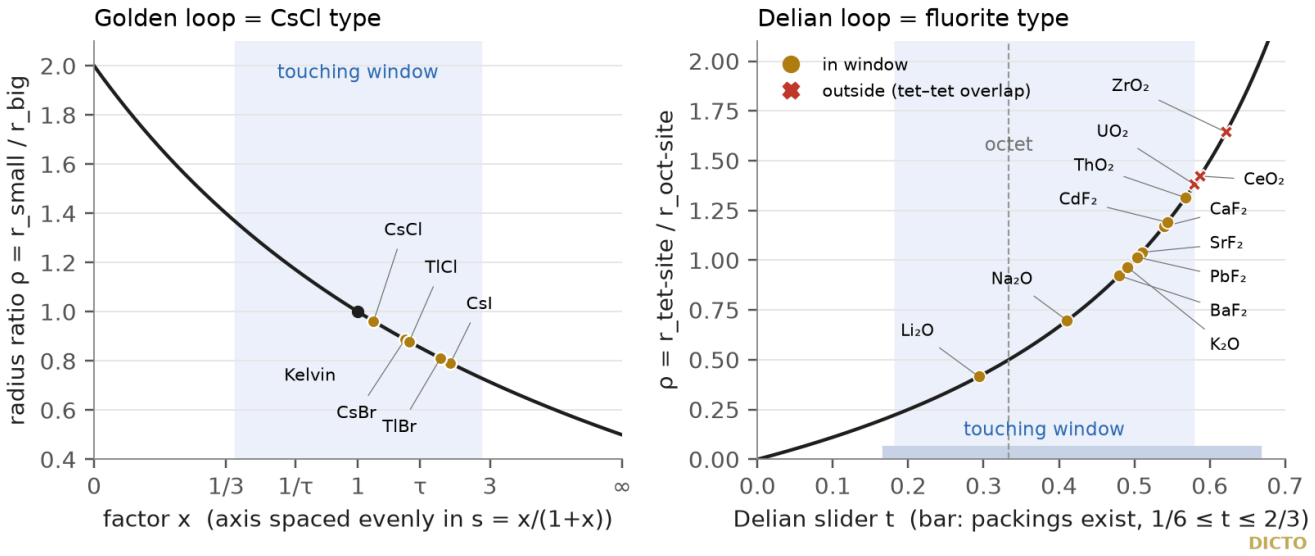


Figure 4. Real compounds on the loops (contact model; Shannon effective ionic radii [8]). Curves: the contact maps of Theorem 2(c). Shaded: the touching windows, whose ends are Pauling's limits. Left: the five caesium chloride type halides lie between Kelvin (x = 1) and the like-contact limit √3 + 1; the golden stop x = τ lies among them. Right: the fluorides and oxides; UO_2, CeO_2 and cubic ZrO_2 fall outside (red crosses). Only Li_2O lies before the octet, and that is not robust (§6).

Agreements with crystal chemistry. The sites are exactly B2 and C1. The touching windows end at Pauling's limits; this is Pauling's own contact condition [9], rederived in the language of radical tessellations. Wang et al. [25] built a binary hard-sphere crystal of large spheres on a simple cubic lattice with a small sphere in every cubic pore (their "SC" crystal: the caesium chloride arrangement at the pore-filling size ratio √3 − 1) and found that the radical polyhedra of both the small and the large spheres have 14 faces, 6 quadrangles and 8 hexagons: that is the golden loop's window end x = √3 + 1, a published point on the loop. The five caesium chloride type halides lie inside the window at x ≈ 1.1–2.1, on the big-ion side of Kelvin, and the golden stop x = τ ≈ 1.618 lies among them (between CsBr/TlCl and TlBr/CsI).

Each golden piece's 14 faces (8 unlike + 6 like) match the two body-centred cubic neighbour shells, 8 at (√3/2)a and 6 at a. Cubic ZrO_2 lies far outside the window ($r^+/r^- = 0.61$), consistent with cubic ZrO_2 needing dopants to be stable at room temperature.

6. Limits of the hard-ball reading

> *The balls are a reading of the geometry, not a model of the crystals. We tested where the reading fails, and why. The answer is mostly the radii, not the physics.*

Three compounds fall outside the touching window with effective ionic radii, all beyond the tet–tet contact edge $(\sqrt{3} + 1)/2 \approx 1.3660$: UO_2 ($\rho = 1.3800$, 1.02 % beyond the edge), CeO_2 (1.4227, 4.15 %) and cubic ZrO_2 (1.6429, 20.27 %), the gap measured as (ρ − edge)/edge; and Li_2O lies on one side of the octet or the other depending on the radius convention [S3]. The misfit is **not an artefact of the contact model**: placing UO_2 and CeO_2 from their measured lattice constants (a = 5.47127 Å [28] and 5.411102 Å [29]) by the power-diagram weight alone, with no contact assumption, gives t = 0.5806 and 0.5878, still beyond $1/\sqrt{3} \approx 0.5774$ [S3, Q1e].

Misfit	Shift needed	Compared with	Verdict
UO_2	+0.0102 Å on U^{4+}, or −0.0140 Å on O^{2-}	additivity residual $r^+ + r^- - d$ = +0.011 Å against the measured lattice [28]; tabulation half-unit 0.005 Å	**within data error**
CeO_2	+0.040 Å on Ce^{4+}, or −0.055 Å on O^{2-}	about 6× its additivity residual (0.007 Å, [29]); crystal radii put it inside	**a radius-convention effect**; no named physical effect (pressure, temperature, gravity, stoichiometry) explains it at room conditions
cubic ZrO_2	+0.170 Å on Zr^{4+}	in yttria-stabilised zirconia the charge-compensating oxygen vacancies sit next to Zr, while Y keeps 8-fold coordination [35]; reaching the window by averaging radii would need a Ca^{2+} cation fraction of 0.61	**explained by defect chemistry**: Zr is not 8-coordinated, so the 8:4 input does not describe the real material
Li_2O (octet side)	+0.12 Å on Li^+, or −0.24 Å on O^{2-}	crystal radii and the Li^+ CN VI radius both cross the octet; pressure cannot decide it: 73.9 GPa would be needed, above the antifluorite → anticotunnite transition at 50 ± 5 GPa [32]	**undecidable from the radii**

Table 4. The misfits, from the scale-anomaly study [S3, Q1, Q4, Q7]. Physical numbers there are estimates and labelled as such.

Pressure, temperature, gravity. In both structures every atom sits on a special position, so uniform compression or expansion scales all lengths alike and **leaves ρ unchanged**; pressure can move ρ only if anion and cation compress by different amounts. Even in the bound most favourable to entering the window (a bookkeeping bound: the whole bond shortening is charged to the anion's hard-sphere radius; $K_0' = 4$ assumed for both, as in [31]), UO_2 would need 3.8 GPa (K_0 = 207 GPa [30]) and CeO_2 18.9 GPa (K_0 = 230 GPa; its fluorite structure transforms at 31.5 GPa [31]); if the cation takes it, both move further out. In any case the radii are fitted to ambient structures, so neither pressure nor temperature can explain an ambient misfit. Heating does not help the reading either: above the superionic transition (for example 1430 K for CaF_2, 1350 K for Li_2O and 2670 K for UO_2, reference values quoted in

[33]) the anion sublattice is highly disordered and the tet-site ball has no fixed site, so the model does not apply; UO_2 expands by 3.39 % from 293 K to 2670 K [34]. Gravity is negligible: a 1 m crystal of UO_2 is strained by its own weight by about 1.7×10^{-7}, and the gravitational force between U^{4+} and O^{2-} is 3.8×10^{-34} of their Coulomb force [S3, Q5].

The biggest effect is the radius convention. Moving from effective ionic radii to crystal radii (each cation +0.14 Å, each anion −0.14 Å) shifts ρ by 21–37 %, more than any physical effect: it puts all three misfits inside the window and moves Li_2O past the octet. Pauling's radius-ratio rule itself is a weak predictor: in about 5000 oxides, George et al. [27] found only 66 % of the tested local environments in agreement with the first rule, and only 13 % of the oxides satisfying the second to fifth rules together; they also note that Shannon radii depend on the environment and cannot be used for local-environment prediction. Shannon's own stated accuracy could not be verified, since the full text of [8] was not read; only the 0.01 Å tabulation step and his reliability keys were used.

Other limits. BaF_2 has $r^+/r^- = 1.08 > 1$, and Na_2O (0.697) and K_2O (0.965) lie far above the 0.414 upper limit for 4-coordination, yet all three are fluorite or antifluorite; the model tests only overlap and has no upper "rattling" limit. The radical faces are not the distance shells: in fluorite with $t > 1/3$ the cation has 12 cation neighbours at $a/\sqrt{2}$ but no cation–cation face. The halide radii are CN VI proxies; the radii come from a transcription of Shannon's table. Left out: NH_4Cl, the metallic B2 alloys and Rb_2O, for lack of verified radii.

7. Conclusion

> *DICTO's web of bonds is a lattice that changes with the loop, as was DICTO's idea: the classical dual of a weighted Voronoi diagram. In the golden loop it changes only in size; in the Delian loop it changes its shape once, at the octet.*

Both DICTONIC loops are one fixed crowd of points with one weight slid between two kinds, and the web of bonds is the dual weighted-Delaunay web. Its rhythm differs between the two loops. In the golden loop the dual never flips: the like bonds trade size in the ratio x^2, equal at Kelvin's cell, and the whole loop is symmetric under $x \leftrightarrow 1/x$. In the Delian loop the dual flips once, at the octet, where every rhombus of a hidden rhombic dodecahedron exchanges its long diagonal for its short one, and the octet is the one member whose web joins only unlike pieces. Read as balls, the loops are the caesium chloride and fluorite structures, and Pauling's classical limits mark where the balls can touch. Open: the scale-anomaly study (§6) and the keyed bond's web. The prior-art search is reported in Appendix A.

8. What is known, and what is new

> *Power diagrams, their duality with weighted Delaunay subdivisions, radical tessellations of ball packings, the two crystal structures and Pauling's rule are all classical. What we did not find is the family between their classical members along these two loops.*

Known. Power (Laguerre, weighted Voronoi) diagrams and their duality with regular triangulations [3, 4]; regular subdivisions change only at the walls of the secondary fan [5, 6], including for periodic decompositions (Alexeev [24, §5.12]); every Delaunay tessellation is the power diagram of its circumspheres [3, 23], so the octet at $t = 1/3$ and the octahedron–cuboctahedron honeycomb at $x = 0, \infty$ are power diagrams with the weights of the Main theorem, and $x = 1$, $t = 0$ and $t = 1$ are ordinary Voronoi diagrams (at $t = 0$ and $t = 1$ of one kind of site alone); radical tessellations of packings of balls of two or more sizes [7], and

Voronoi–Dirichlet and radical polyhedra in crystal chemistry and coordination [18, 19, 21]; two unequal truncated octahedra on a body-centred lattice as Laguerre cells (general; noted in paper #48 [1]); the octet, rhombic dodecahedron and cube connections [11]; the T(x) orbit [10]; ionic radii [8] and the radius-ratio rule [9]; the window ends are Pauling's contact limits [9]; radius-ratio-dependent packings of two sphere sizes (Murray and Sanders [26]); a binary hard-sphere crystal of large spheres on a simple cubic lattice with a small sphere in every cubic pore, whose radical polyhedra all have 14 faces (Wang et al. [25], their "SC" crystal; in our reading the caesium chloride arrangement at size ratio $\sqrt{3} - 1$); caesium chloride type and other binary crystals of colloids and nanoparticles whose phase is set by the size ratio [12–17]. Paper #51's cube Remark [2] and its octet check already gave the two weights at samples. Voronoi's theory of Delaunay subdivisions of lattices (Erdahl [22]) is related, but treats lattices, not two kinds of weighted sites.

Not found, in the sources listed in Appendix A: the closed-form bond areas of both loops; the contrast between the golden loop (no flip) and the Delian loop (one flip, at the octet); the like-bond ratio x^2 and the diagonal sum φ; the sequence of dual cells between the classical end and middle members; the octet as the only member with a bipartite web; the contact maps $\rho = (x + 2)/(2x + 1)$ and $\rho = t/(1 - t)$; the Delian packing range $1/6 \le t \le 2/3$. These are elementary consequences of standard theory and may well be known in some form; "not found" is limited to those sources, which include the prior-art search of 25 October 2026 summarised in Appendix A.

9. In the app

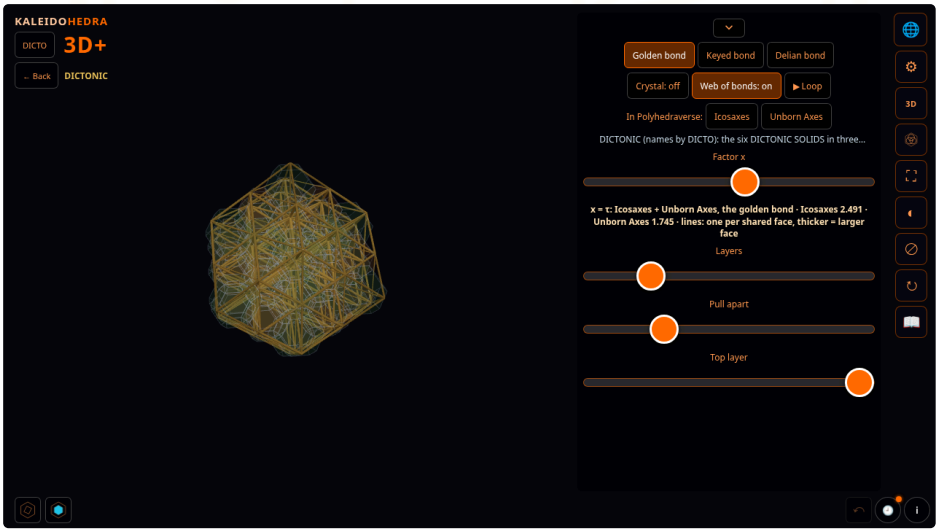


Figure 5. App screenshot (Kaleidohedra, DICTONIC world, golden bond at $x = \tau$, "Web of bonds: on"; https://dictospheres.com): one line per shared face, thicker for a larger face. The ▶ Loop button runs the factor through the loop, and the web changes with it.

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[S2] DICTO, physical anchors study (studies/2026-10-25-physical-anchors, commit 28cfdde): scripts/symbolic.py → symbolic-checks.txt (part A, 32 checks), scripts/compounds.py → compounds-checks.txt and data/raw/compounds.json (part B, 29 checks), scripts/crossref_check.py → data/raw/crossref.json.
[S3] DICTO, scale-anomaly study (studies/2026-10-25-scale-anomalies, commit 28cfdde): scripts/anomalies.py → data/checks/anomalies-checks.txt (24 checks; Q1 error bars, Q2 pressure, Q3 temperature, Q4 metastability, Q5 gravity) and data/raw/values.json (every external number, with how its source was read).

Appendix A. Sources checked for prior work

DICTO's reference library (kaleidohedra-references): nine ask-library queries (power diagram / Laguerre / weighted Voronoi; contact graph of a space-filling; the Koca truncated-octahedron family on a body-centred lattice; the octet / rhombic dodecahedron / cube family; regular triangulations, flips and the secondary polytope; caesium chloride type binary colloidal crystals with tunable size ratio; radical tessellation and Laguerre–Voronoi polyhedra of sphere packings in crystals; ionic radii, radius ratio, coordination and Pauling's rules; the fluorite structure and the tetrahedral holes of the face-centred cubic lattice), and its index, reading list and README searched by keyword. Nothing was found there on power diagrams, Laguerre or radical tessellations, contact graphs, secondary polytopes, ionic radii or radius-ratio rules; the nearest entries were Koca et al. 2015 [10], Erdahl 1999 [22] and Garber 2019 (Voronoi's theory for lattices), and Cadotte, Dshemuchadse, Damasceno, Newman and Glotzer 2016 (binary hard tetrahedra and octahedra, not balls). Web, 25 October 2026: one search for radical tessellation in ionic crystals, which found no paper linking power diagrams with ionic radii, caesium chloride or fluorite. References [3–5, 7–9, 12–19, 21, 22]: every DOI checked on Crossref (title, journal, year, volume and pages); none of these papers was read in full for this paper. Reference [11] is cited as in paper #51. **Prior-art search, 25 October 2026** (file prior-art.md in this paper's folder): four further ask-library queries (power diagram / Laguerre and radius ratio; radical Voronoi binary packing CsCl; lattice Delaunay subdivisions, L-types and the secondary fan; Laguerre weight flips), still with no power-diagram, radical-tessellation or ionic-radius paper in the library; about 15 web queries (radical, power and Laguerre tessellation with caesium chloride, fluorite, ionic radii, radius ratio, binary hard-sphere crystals, bidisperse foams, Fischer–Koch, Blatov / ToposPro, periodic secondary fans and Delone subdivisions, Delaunay as a power diagram); the full text of Alexeev [24] searched for its periodic sections; Crossref for every DOI, including [6], [20] and [23–27]. Also identified: Rybnikov 1999 (liftings of cell complexes), Richard, Oger, Troadec and Gervois 1998 (radical tessellation of binary random assemblies), Koch 1973 (Dirichlet domains of cubic lattice complexes), Erdahl and Rybnikov (L-types as secondary cones), Schürmann 2010, Hopkins, Stillinger and Torquato 2012. Identified but not read in full: Aurenhammer's 1987 paper on criteria for power diagrams (Discrete and Computational Geometry 2, 49–64), Blatov 2004 [18], Koch 1973, Richard et al. 1998. An independent adversarial review of an earlier version (v2) (referee report review-fable.md in this paper's folder) recomputed all formulas, counts, dual cells, windows, stops and table values with its own code, and checked every DOI; no changes to numbers or theorems resulted, only to wording and scope. Not reached: Google Scholar itself, and closed-form face-area tables in crystal-chemistry handbooks. For §6, references [28, 30–32, 35] were read in their abstracts, [27, 33]

and [34] in full text or saved excerpt, [29] in its table; facts from search summaries only were left out or labelled [S3].

Appendix B. Checks

Study	Part	Checks	PASS	FAIL
web of bonds [S1]	A symbolic, whole regimes (A1–A7)	piece = power cell, areas, volumes, face to face, monotonicity, distance bound	55	0
	B exact samples (B1–B8)	cells from scratch at 10 golden and 11 Delian parameters, rational and irrational	211	0
	C app cross-check (C1–C4)	frame, every app bond area, counts, float findings	105	0
physical anchors [S2]	A symbolic (A1–A7)	weight = r^2, maps, windows, feasibility, stops, B2/C1, area identity	32	0
	B exact cells (B0–B3)	Kelvin and rhombic dodecahedron sanity, 9 loop stops, 17 compounds, Pauling = window	29	0
scale anomalies [S3]	E exact edges, Q1–Q5	window edges, error bars, pressure and temperature bounds, metastability, gravity (physical numbers are estimates)	24	0
total			**456**	**0**

Exact arithmetic: Python fractions over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{6})$, and sympy for everything that depends on the parameter; a number is zero only if all its field coefficients are; floats only choose candidate plane triples and serve as the app cross-check. Golden samples x = 0, 1/3, 1/2, 1, 2, 3, τ, $1/\tau$, $\sqrt{2}$, ∞; Delian samples t = 0, 1/6, 1/4, 1/3, 1/2, 2/3, 1 and the stops $(2\sqrt{6}-3)/15$, $1-\sqrt[3]{6}/3$, $1-(2/3)\cdot 2^{-1/3}$, $5/3-(2/3)\sqrt{2}$.